PCSrcE

PCSrcE = PCSrcE0 | PCSrcE1

Issue0, Issue1: CLR de ID/EX por vía

PCSrcE0 = {JalrE0, BranchE0·ZeroE0 + JumpE0}

PCSrcE1 = {JalrE1, BranchE1·ZeroE1 + JumpE1}

Issue Unit

RawPair

MemConflict

CtrlConflict

LoadUse0 / LoadUse1

Issue0 = Valid0 ·~LoadUse0

Issue1 = Issue0 & Valid1 & ~LoadUse1 & ~RawPair & ~MemConflict & ~CtrlConflict

Consumed

Control Unit 0

op 6:0

funct3 14:12

$funct7_5$ 30

tipo, rd, rs

RegWriteD0

ResultSrcD0

MemWriteD0

JumpD0

BranchD0

JalrD0

ALUControlD0

ALUSrcD0

Control Unit 1

op 6:0

funct3 14:12

$funct7_5$ 30

RegWriteD1

ResultSrcD1

MemWriteD1

JumpD1

BranchD1

JalrD1

ALUControlD1

ALUSrcD1

ImmSrcD0

Extend 0

31:7

ImmExtD0

PCD0

PCPlus4D0 = PCD0+4

RdD0

11:7

ImmSrcD1

Extend 1

ImmExtD1

PCD1 = PCD0+4

PCPlus4D1 = PCD0+8

RdD1

PC

PCF

00

01

10

A

Instruction Memory

RD

lee PC y PC+4

InstrF

InstrD1

10

01

InstrD0

Valid0

Valid1

RD1

{Consumed,00}

+

PCSeqF

IF/ID

slot 0, slot 1 y valid

CLR = FlushD · se congela si Consumed = 00

WE3

WE3

A1 19:15

A2 24:20

A3

WD3

Register File

4 lecturas · 2 escrituras

si escriben igual, gana la vía 1

RD1

RD2

B1 19:15

B2 24:20

B3

WDB3

RDB1

RDB2

v0

RegWriteE0

ResultSrcE0

MemWriteE0

JumpE0

BranchE0

JalrE0

ALUControlE0

ALUSrcE0

ZeroE0

ImmExtE0

PCE0

RD1E

RD2E

000

001

010

011

100

SrcAE0

SrcBE

0

1

ALU 0

ALUResultE0

WriteDataE0

PCPlus4E0

RdE0

RegWriteE1

ResultSrcE1

MemWriteE1

JumpE1

BranchE1

JalrE1

ALUControlE1

ALUSrcE1

InstrF1

InstrD1

ImmSrcD1

ImmExtE1

PCE1

SrcAE1

SrcBE

ALU 1

ALUResultE1

WriteDataE1

PCPlus4E1

RdE1

IDEX

CLR v0 = FlushE + ~Issue0 ·

CLR v1 = FlushE + ~Issue1

RegWriteM0

ResultSrcM0

MemWriteM0

ALUResultM0

WriteDataM0

PCPlus4M0

RdM0

JPCSrcE1

TakenL1

ZeroE1

MemSelM2

01

WE

A

Data Memory

1 puerto

RD

WD

ReadDataM

RegWriteM1

ResultSrcM1

MemWriteM1

ALUResultM1

WriteDataM1

PCPlus4M1

RdM1

EX/MEM

CLR v1 = FlushM1 (v0 sin clear)

RegWriteW0

ResultSrcW0

RdW0

ResultW0

00

01

10

RegWriteW1

ResultSrcW1

RdW1

ResultW1

MEM/WB

PCTargetE

{ALUResultE[31:1], 0} (jalr)

ResultW0

ResultW1

ALUResultM0

ALUResultM1

RdW0

RdW1

FlushD

LoadUse0, LoadUse1

FlushE

Rs1E0, Rs2E0, RdE0

Forward(A0, B0, A1, B1

Rs1E1, Rs2E1, RdE1

FlushM1 (vía 1)


M2
M0 M1 W0 W1
Hazard Unit
Entradas de ambas vías: Rs1D, Rs2D, UsesRs (ID) · Rs1E, Rs2E, RdE, LoadE, PCSrcE (EX) · RdM, RegWriteM · RdW, RegWriteW
Forward por operando: M1 > M0 > W1 > W0 > registro · FlushD = FlushE = taken0 + taken1 · FlushM1 = taken0 · LoadUse0/1 a la Issue Unit
Vía 0
Vía 1
Control compartido
Forwarding